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The Protocol Stack

2.3 Other relevant protocols

Sergi Abadal (abadal@ac.upc.edu)

Xarxes i Jocs On-line – 2025-2026
Grau en Disseny i Desenvolupament de Videojocs

Syllabus

1. Introduction
2. The protocol stack (3 sessions)
   - Introduction and upper layers
   - Lower layers: network, link, physical
   - **<u>Other relevant protocols</u>**
   - Putting it all together: Wireshark
3. Impact on programming of on-line games
4. Other stories

Remember from last class

- Link layer:
  - Puts bits into frames so that messages make sense
  - MAC protocols to avoid collisions
  - Management and correction of errors with codes
- Physical layer:
  - Need to define how information is put into cables or the air
  - We need ways to avoid and correct errors
- You cannot control link and physical layers, but they affect the delays very much. That’s the difference between mobile and PC gaming

Today

- There are many protocols relevant to gaming, both in the background and in the foreground.
  - *FTP, SMTP, POP3, IMAP, SSH, NFS…*
- Today we will explain some of them, very briefly
  - For connectivity
  - For security
  - For streaming
  - For other stuff

RTP DNS HTTP	7	Application
	6	Presentation
TLS QUIC	5	Session
	4	Transport
NAT ICMP	3	Network
	2	Data Link
	1	Physical

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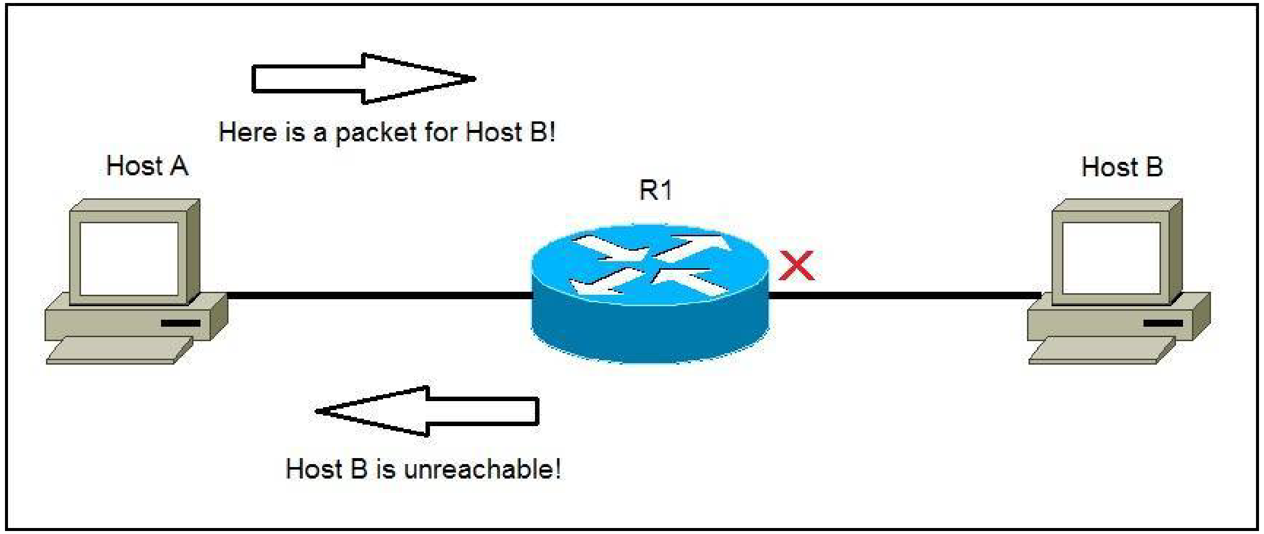
Protocols for connectivity: ICMP and NAT

Session 4: Practical NAT concepts

ICMP: Internet Control Message Protocol

Layer: Network

- Allows routers to send error or control messages to other routers or nodes. Communicates the IP software of several machines.
  - Destination unreachable
  - Echo request/reply (ping)
  - TTL expired
  - ...



ICMP: Internet Control Message Protocol

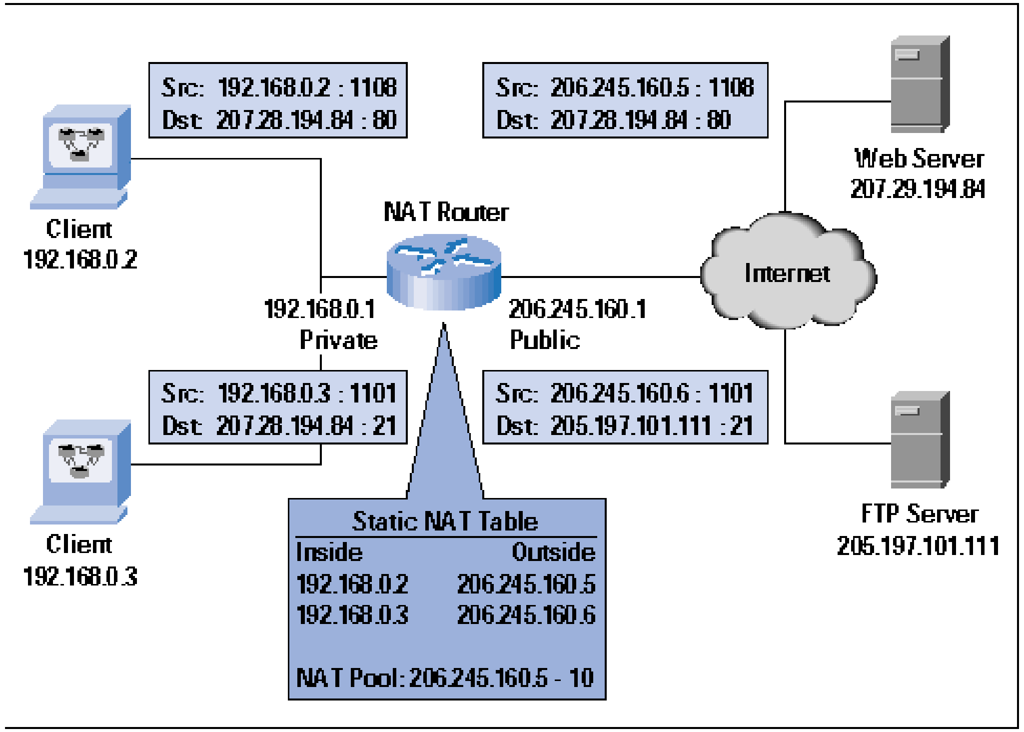
It is important for on-line games because

- It gives you information about the latency and TTL.
- It helps you determine if a connection is alive and the destination is responding (and hence you will use it to implement virtual UDP connections)
- Your game may implement some adaptive process depending on the latency, TTL, and others.

NAT: Network Address Translation

Layer: Network*

- NAT maps IP addresses of two networks. The most extended example is the mapping between the local IPs in a LAN and the Internet. Another example would be translation from IPv4 to IPv6



NAT: Network Address Translation

Multiplayer games

From the lab sessions

- ~~Same machine -> Limited number of players~~
- ~~Local Network -> Need of being physically there~~
- 1 Server + N Clients -> Possible Overload on Server (Multiple game instances?)
- P2P -> NAT (and peer machine dependence)

NAT: Network Address Translation

UPC CITM

Importance of NAT

From the lab sessions

Pros:

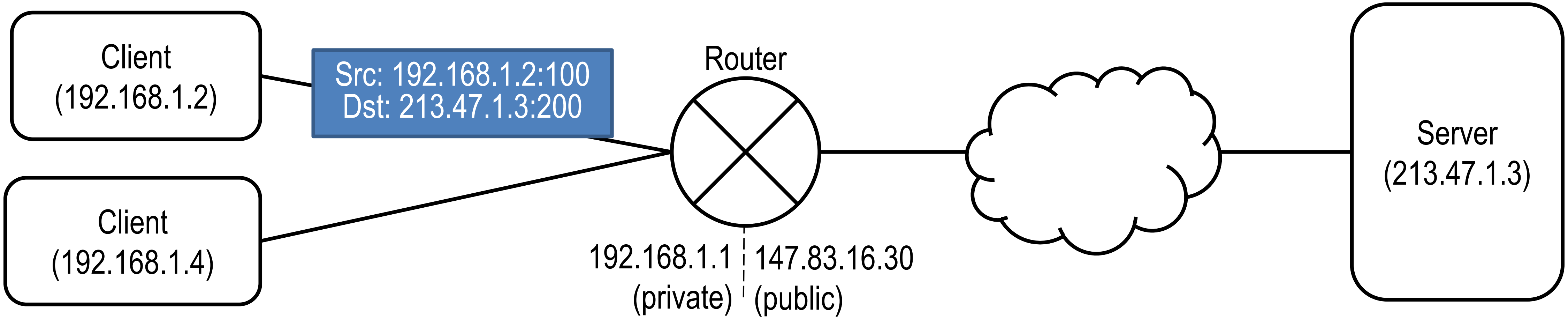
- It gives you a unique identity in the internet
- It protects you from unwelcome connections

Cons:

- It forbids connections from clients unknown to the NAT
- No clear standard. Different clients have different NAT implementations that require different “tricks” to bypass them

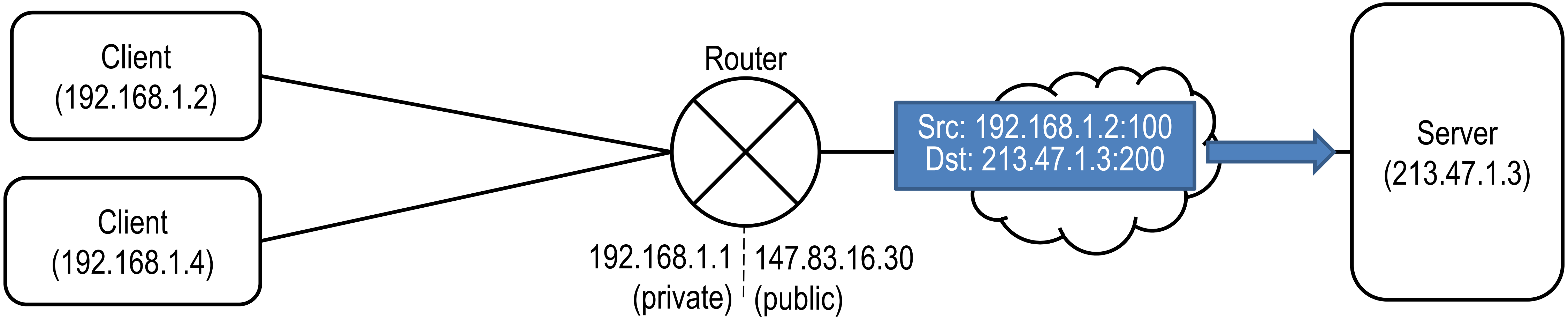
NAT: Network Address Translation (I)

- **Router without NAT**
  - Packet is sent to server



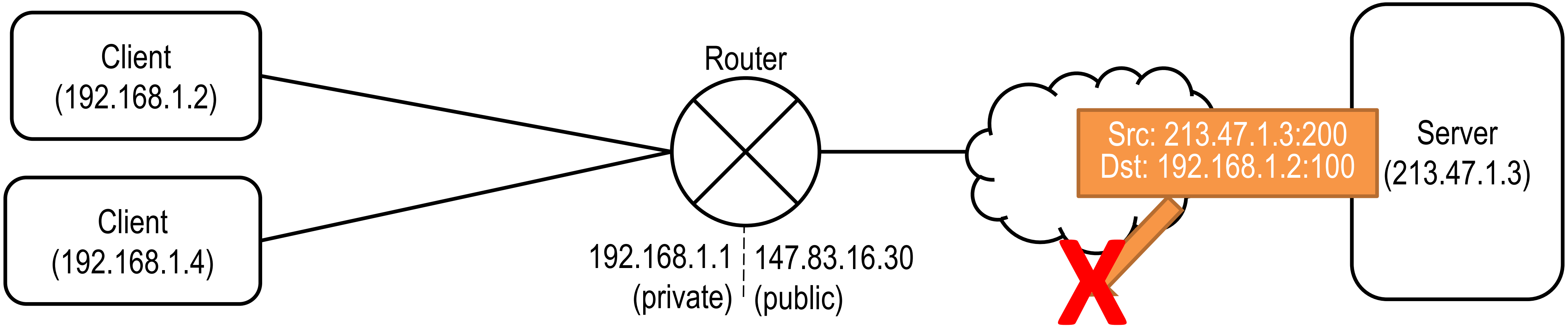
NAT: Network Address Translation (I)

- **Router without NAT**
  - Packet is sent to server
  - Router is capable of reaching the server



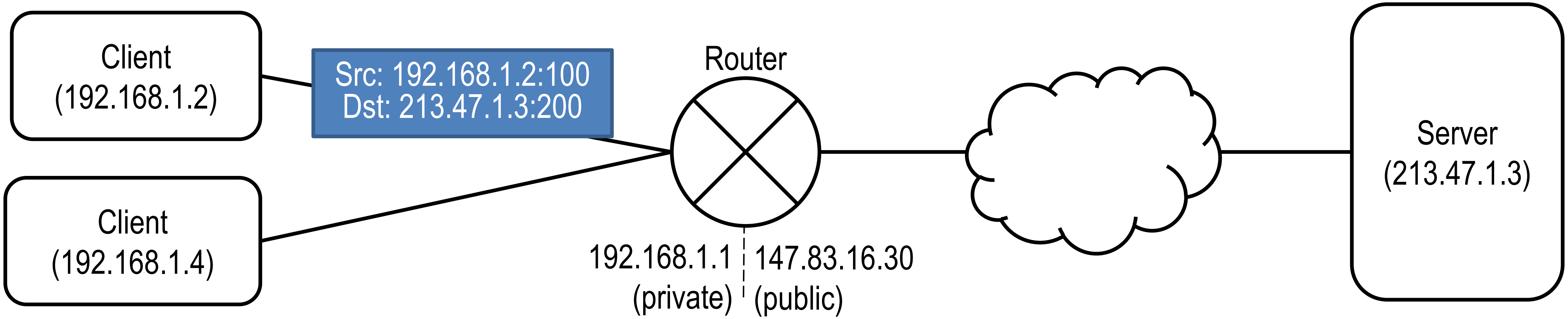
NAT: Network Address Translation (I)

- **Router without NAT**
  - Packet is sent to server
  - Router is capable of reaching the server
  - Server cannot reach the client (private address)



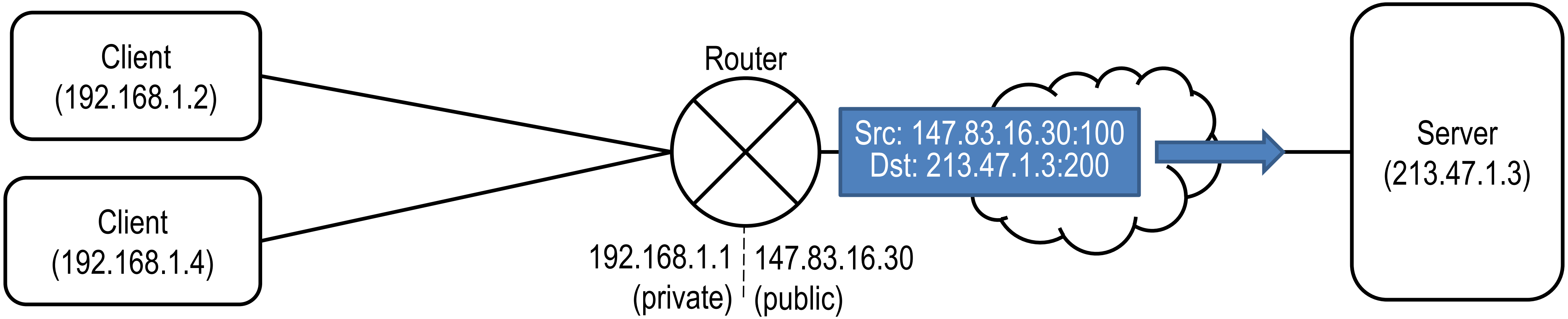
NAT: Network Address Translation (II)

- **Router with naïve NAT**
  - Packet is sent to server



NAT: Network Address Translation (II)

- **Router with naïve NAT**
  - Packet is sent to server
  - Source address is rewritten at NAT router

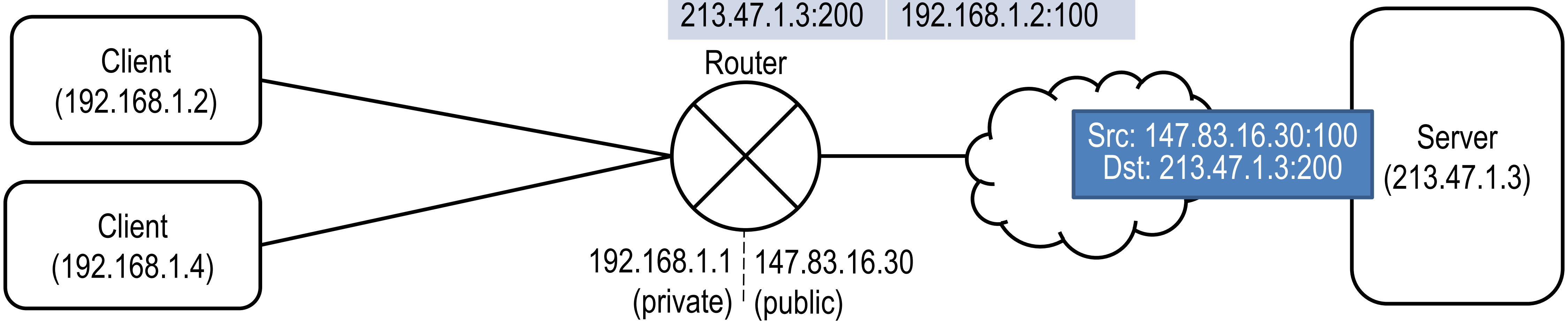


NAT: Network Address Translation (II)

- **Router with naïve NAT**
  - Packet is sent to server
  - Source address is rewritten at NAT router
  - NAT table is filled with new entry

NAT Table

Packets from	Shall go to
213.47.1.3:200	192.168.1.2:100



NAT: Network Address Translation (II)

- **Router with naïve NAT**
  - Packet is sent to server
  - Source address is rewritten at NAT router
  - NAT table is filled with new entry
  - Destination address rewritten

NAT Table

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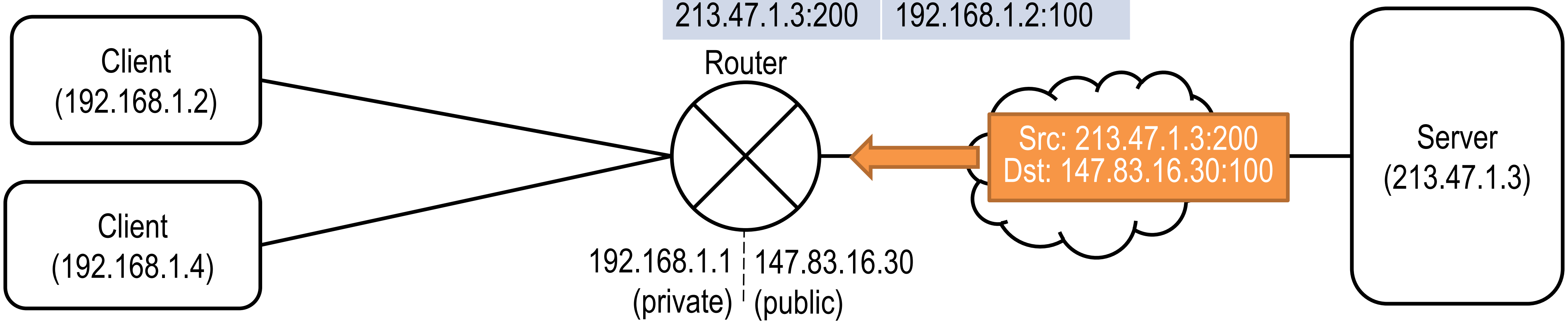
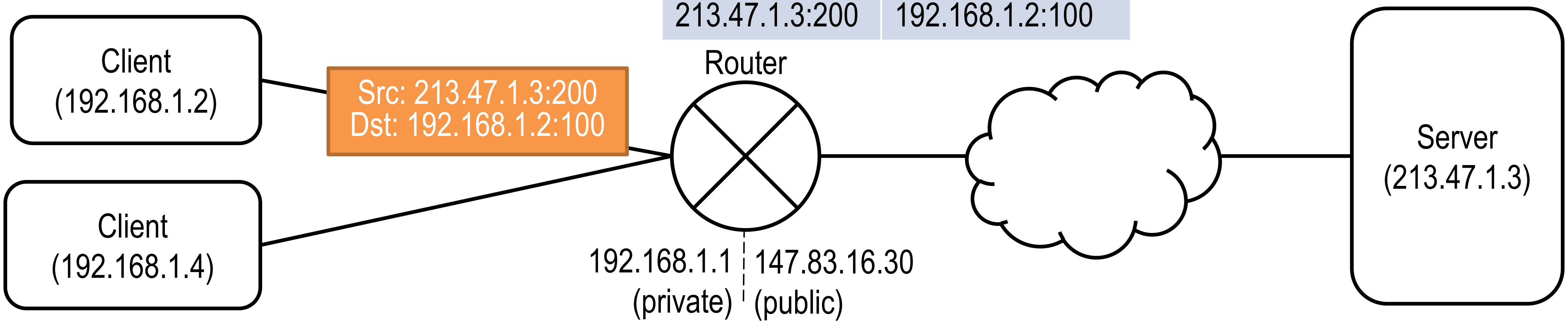


NAT: Network Address Translation (II)

- **Router with naïve NAT**
  - Packet is sent to server
  - Source address is rewritten at NAT router
  - NAT table is filled with new entry
  - Destination address rewritten
  - Reply reaches client

NAT Table

Packets from	Shall go to
213.47.1.3:200	192.168.1.2:100



NAT: Network Address Translation (II)

- **Router with naïve NAT**
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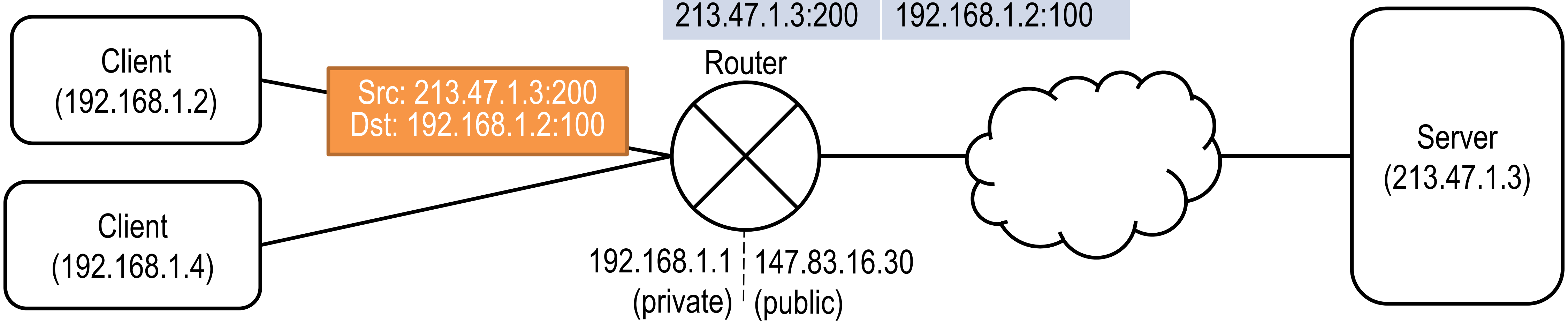
PROBLEM?

NAT Table

Packets from	Shall go to
213.47.1.3:200	192.168.1.2:100



NAT: Network Address Translation (II)

- Router with naïve NAT does not work if two clients from the same private network want to communicate with the same server.
- The response will not know which of the clients to reach, as the NAT table would have duplicate entries.
- Actual NAT protocols assign random ports to each communication and more sophisticated tables

Naïve NAT Table

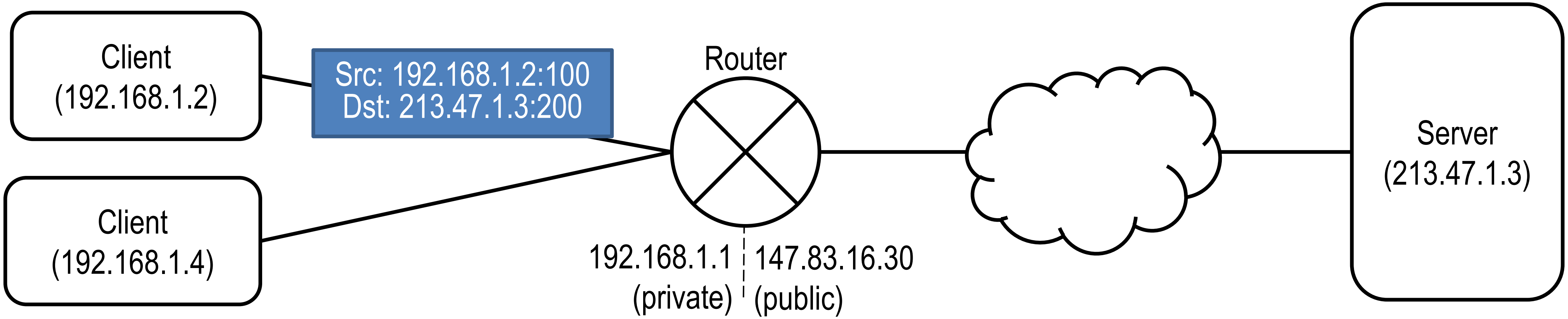
Packets from	Shall go to
213.47.1.3:200	192.168.1.2:100
213.47.1.3:200	192.168.1.4:100

Actual NAT Table

Packets from	External Port	Shall go to
213.47.1.3:200	50000	192.168.1.2:100
213.47.1.3:200	45300	192.168.1.4:100

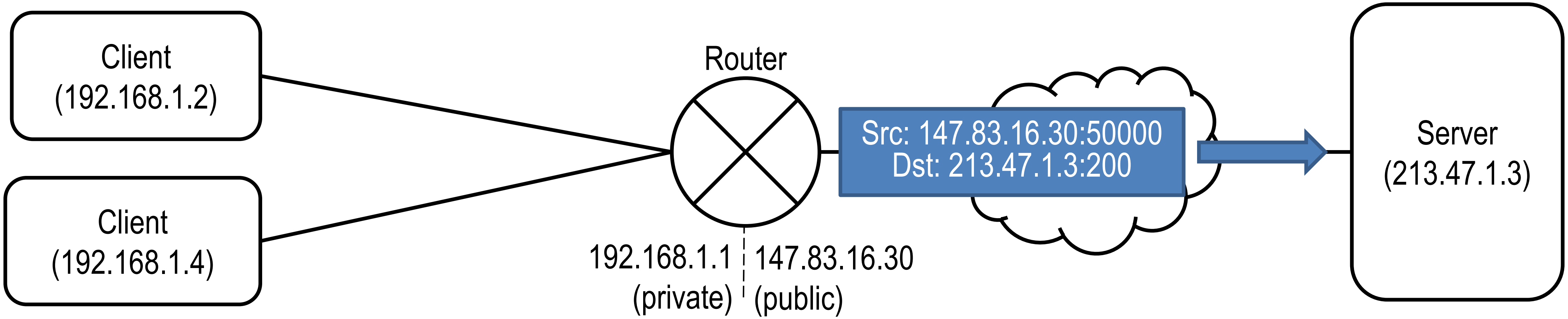
NAT: Network Address Translation (III)

- **Router with actual NAT**
  - Packet is sent to server



NAT: Network Address Translation (III)

- **Router with actual NAT**
  - Packet is sent to server
  - Source address is rewritten at NAT router with random port

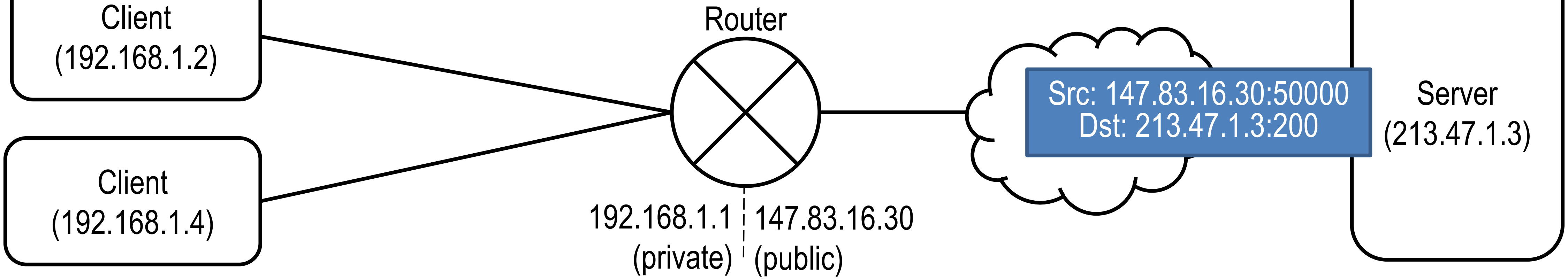


NAT: Network Address Translation (III)

- **Router with actual NAT**
  - Packet is sent to server
  - Source address is rewritten at NAT router with random port
  - NAT table registers this entry with this external port

NAT Table

Packets from	External Port	Shall go to
213.47.1.3:200	50000	192.168.1.2:100



NAT: Network Address Translation (III)

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  - Packet is sent to server
  - Source address is rewritten at NAT router with random port
  - NAT table registers this entry with this external port
  - Destination address rewritten

NAT Table

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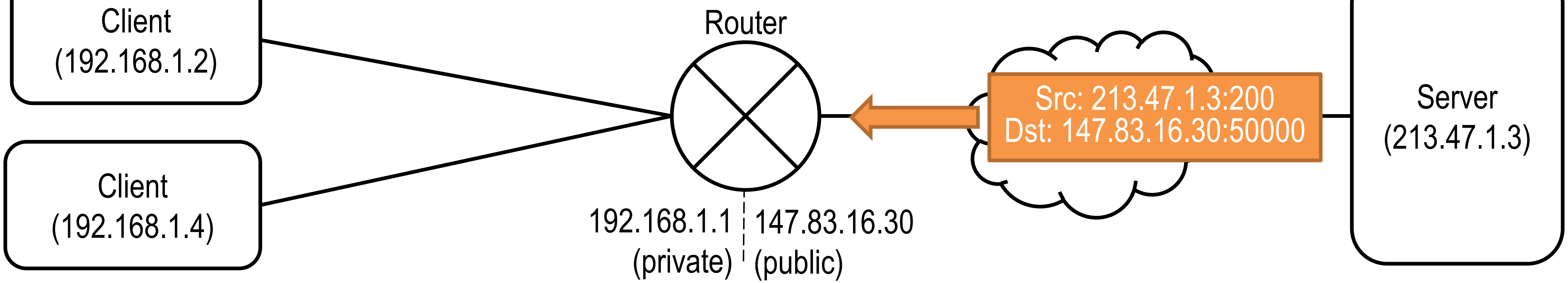

NAT: Network Address Translation (III)

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NAT Table

Packets from	External Port	Shall go to
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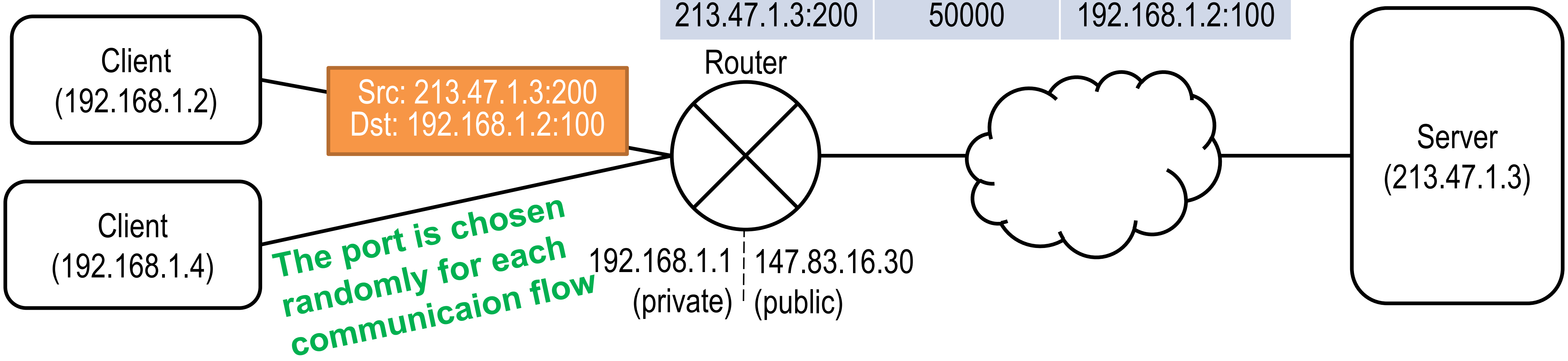


NAT: Network Address Translation (III)

- The NAT would work even if the table did not contain the “packets from” field (just by associating external ports to a private address)

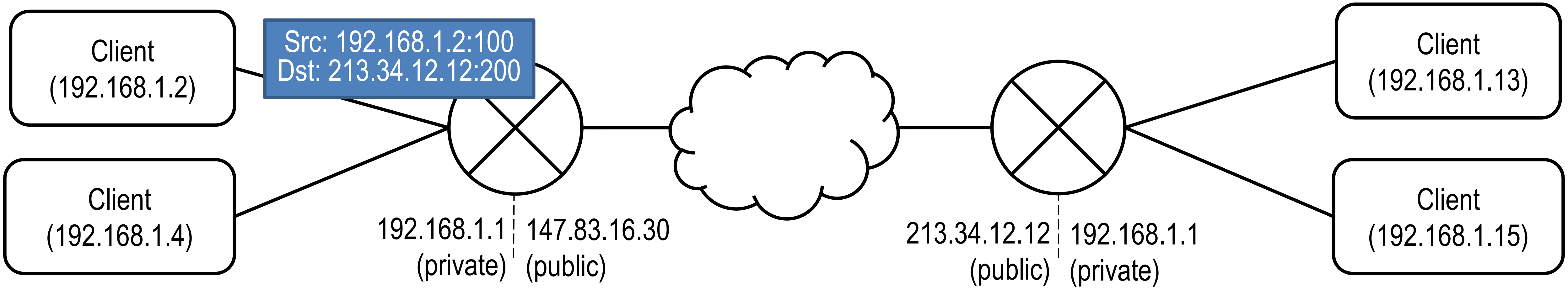
Actual NAT Table

Packets from	External Port	Shall go to
213.47.1.3:200	50000	192.168.1.2:100
213.47.1.3:200	45300	192.168.1.4:100

- However, this is included for security (to not allow external entities access private addresses by trying random ports).
- Source and external port need to match. If only the port matches, then packets are discarded.
- **This creates problems in some cases relevant to gaming.**

NAT Traversal

- The problem is called NAT Traversal: how do you reach a client which is “behind NAT” from a location that is “behind NAT” as well?
  - The source will go past the first router, but not the last



NAT Traversal

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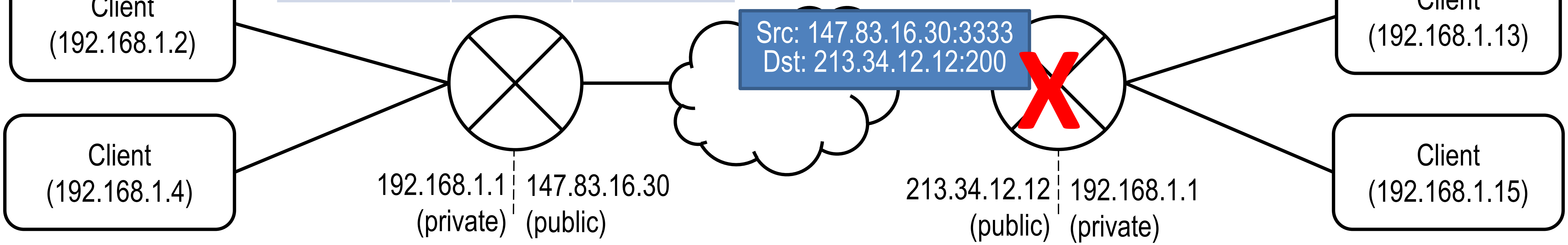


Packets from	Ext. Port	Shall go to
213.34.12.12:200	3333	192.168.1.2:100

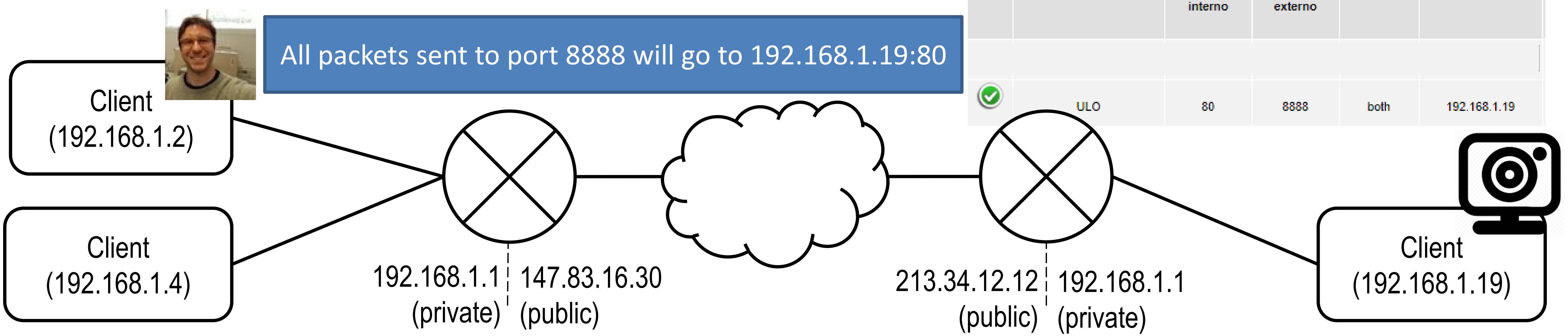
Packets from	Ext. Port	Shall go to
-	-	-

NAT Traversal

- The problem is called NAT Traversal: how do you reach a client which is behind NAT from a location that is behind NAT as well?
  - **First solution: change router properties (manual)**

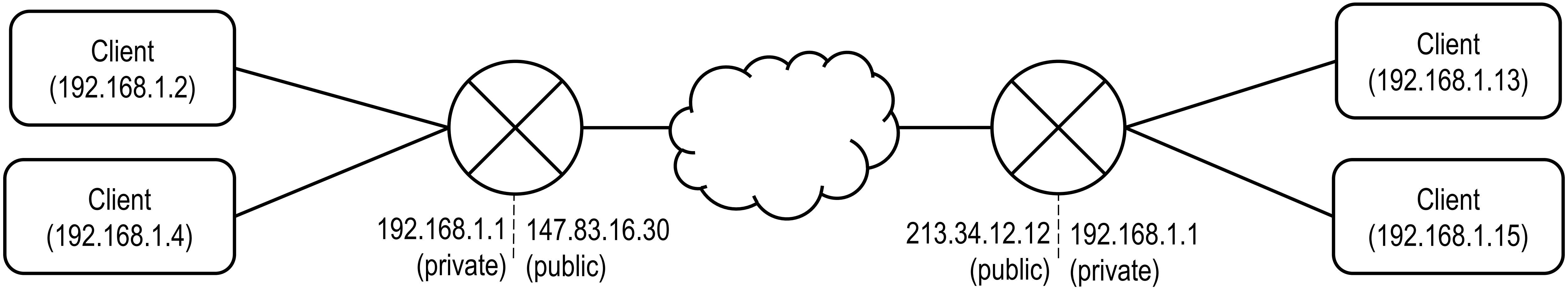
Personalizar reglas

estado	aplicación / servicio	puerto interno	puerto externo	protocolo	IPv4 del dispositivo
✓	ULO	80	8888	both	192.168.1.19

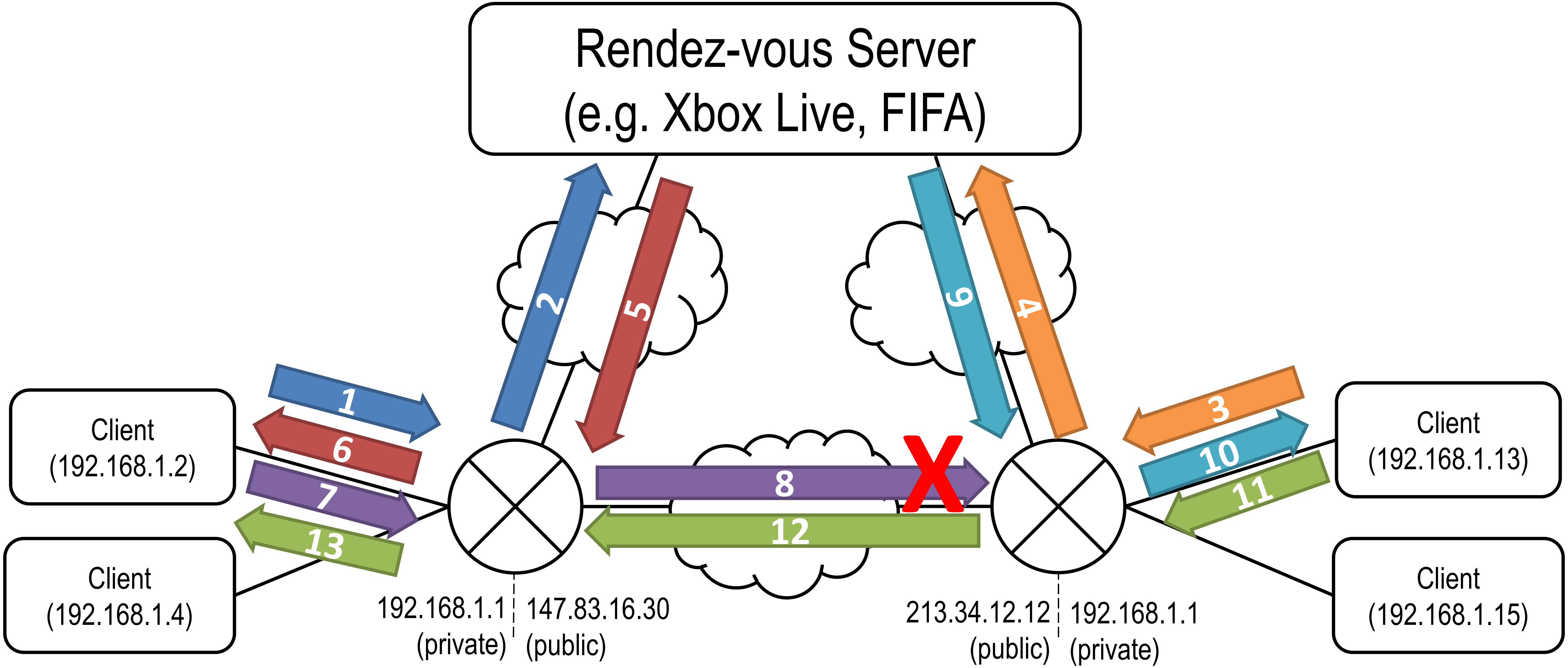


NAT Traversal

- The problem is called NAT Traversal: how do you reach a client which is behind NAT from a location that is behind NAT as well?
  - First solution: change router properties
  - **Second solution: hole punching (automatic)**

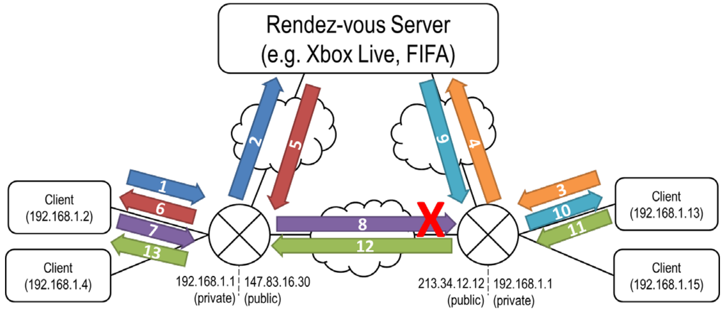


NAT Traversal: Hole Punching



NAT Traversal: Hole Punching

- Client1 requests connection to server. NAT table1 now contains server address.
- Client2 requests connection to server. NAT table2 now contains server address.
- Server sends IP of client2 to client1.
- Client1 tries to communicate with client2. Message does not arrive, but NAT table1 now contains client2 address.
  - In the meanwhile, server notifies client2 telling him that client1 is ready to receive his messages.
- Client2 tries to communicate with client1. NAT table2 now contains client1 address. Since NAT table1 already contains client2 address (previous step), this is successful.



NAT Traversal

- The problem is called NAT Traversal: how do you reach a client which is behind NAT from a location that is behind NAT as well?
  - First solution: change router properties
  - Second solution: hole punching
  - **Third solution: use ICE/STUN/TURN protocols (automatic)**

NAT Traversal: There's more…

CITM

From the lab sessions

NAT Traversal

Hole Punching -> Send a message (lost) to an "unknown" client to tell the NAT that it should allow incoming messages

STUN servers -> Servers dedicated exclusively to share your public IP and routing port to kown clients

https://en.wikipedia.org/wiki/STUN

NAT types -> (I, II, III, full cone, symmetric, etc.) There are multiple types of NAT, and Hole punching will not work in all of them

Port prediction -> Some times, ports used can be predicted in the most restrictive cases

TURN servers -> relay information from clients (more load than STUN, less load than dedicated server)

https://en.wikipedia.org/wiki/Traversal_Using_Relays_around_NAT

- The Interactive Connectivity Establishment (ICE) technique uses STUN/TURN.

https://www.tutorialspoint.com/understanding-interactive-connectivity-establishment-ice-a-comprehensive-guide

NAT: Network Address Translation

It is important for on-line games because

- You may need to define different methods to connect to the server and remote client depending on the type of NAT supported
  - That's why some games allow to change this in the settings
- You may need to decide whether you want to implement your own *rendez-vous* server, build a STUN/TURN server, or buy STUN/TURN services from a third party.

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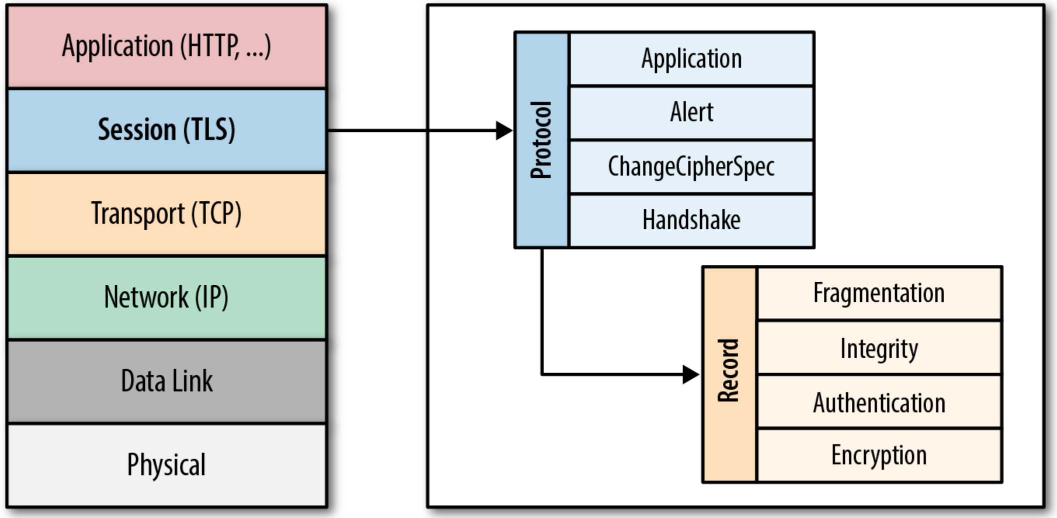
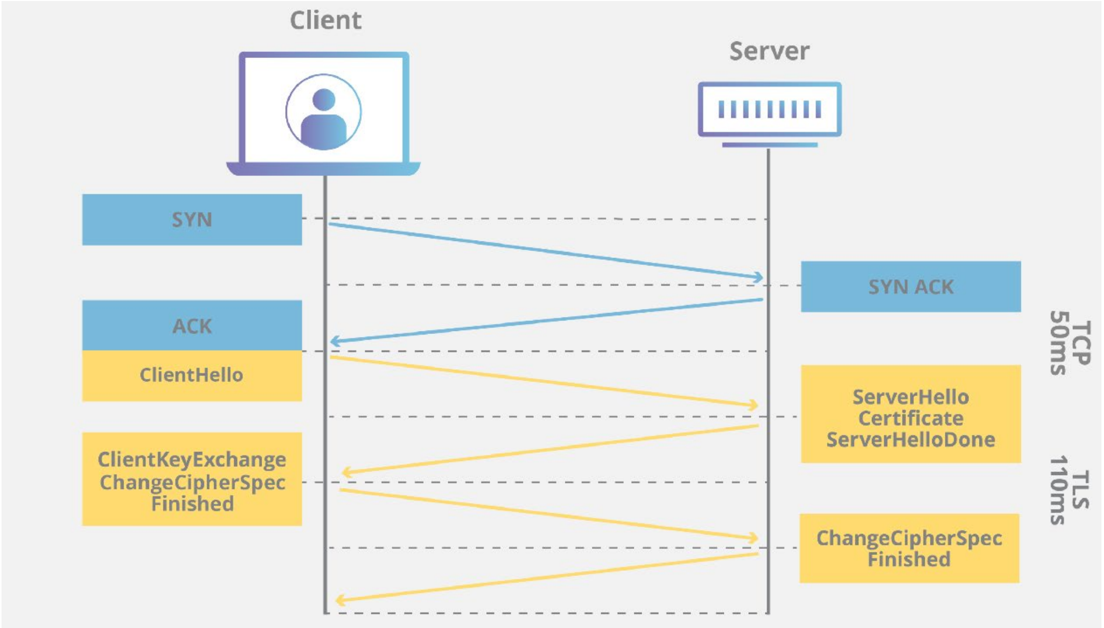
Protocols for security: TLS and QUIC*

TLS: Transport Layer Security

Layer: Session/Transport

- Provides security against eavesdropping and tampering
- Does it through a handshake to share unique keys, which are use to encrypt the data
- The server has a certificate issued by a trusted party
- Works over TCP as it needs someone to handle losses and reordering

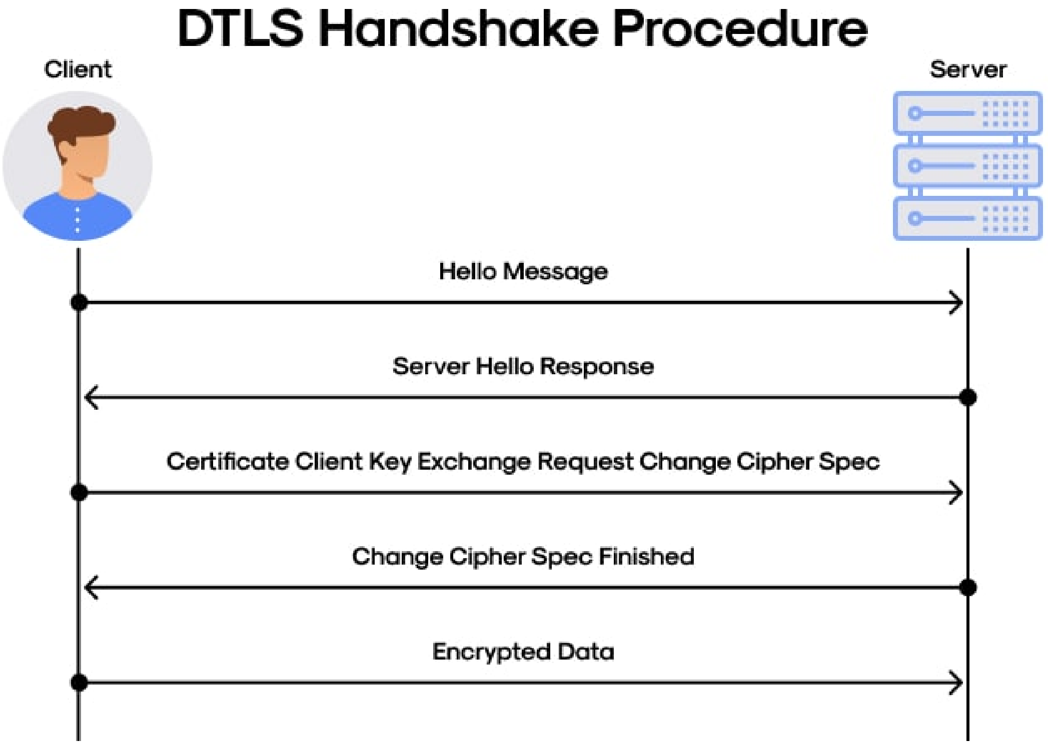




DTLS: *Datagram* Transport Layer Security

Layer: Session/Transport

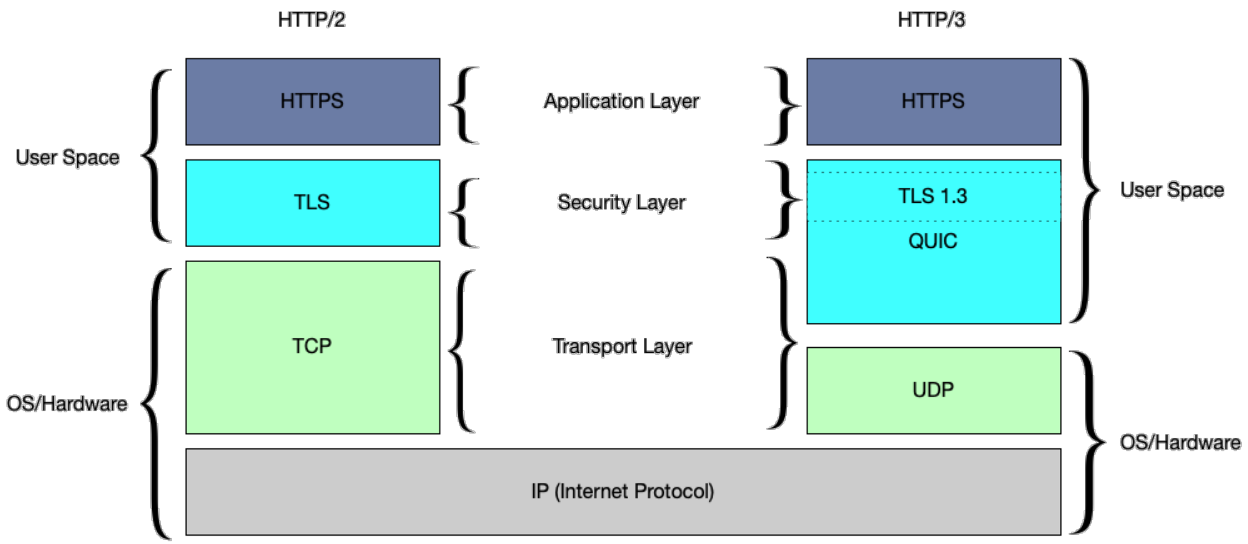
- TLS requires TCP, which is bad for latency. People wanting to use UDP (like us) need an alternative.
- DTLS provides equivalent security guarantees than TLS, but results in lower latency.
- It would be still be better to have a new transport protocol that inherently includes security…

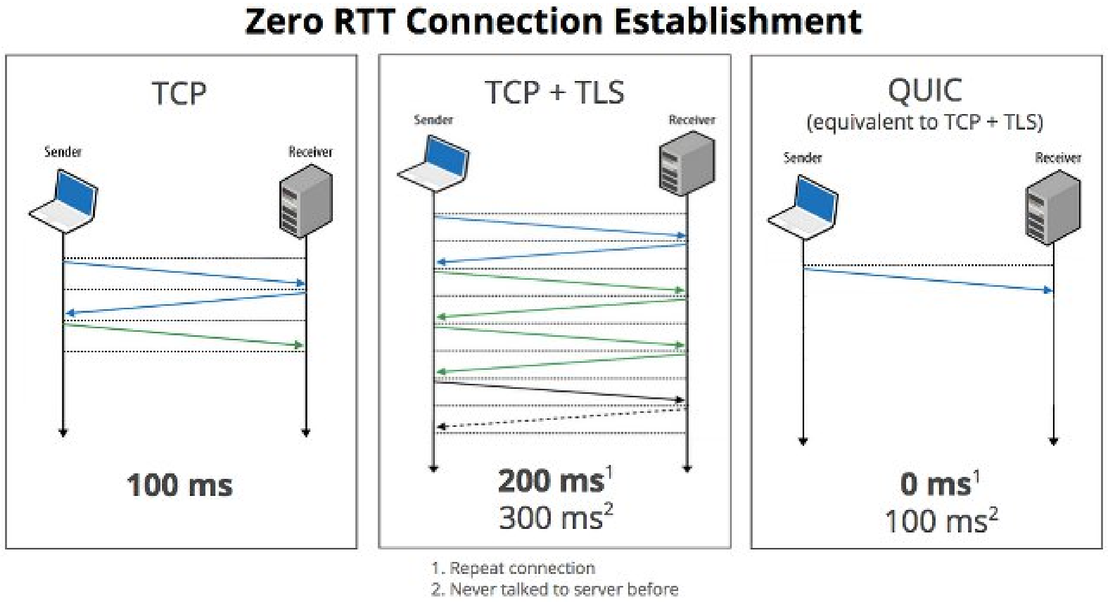


QUIC: Quick UDP Internet Connections

Layer: Transport*

- Proposed by Google as a replacement of TCP
- Technically it is a protocol built over UDP.
- Key traits:
  - Offers security through TLS
  - Implements reliability through several configurable methods
  - Data remains at the user space, unlike TCP
  - All of this, faster than TCP





QUIC: Quick UDP Internet Connections

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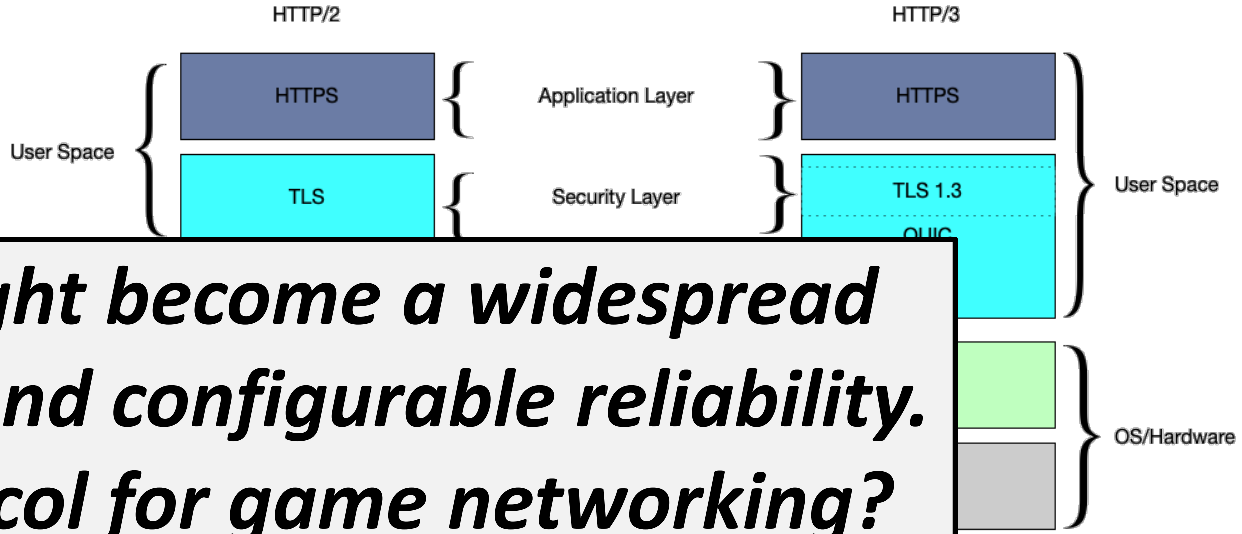
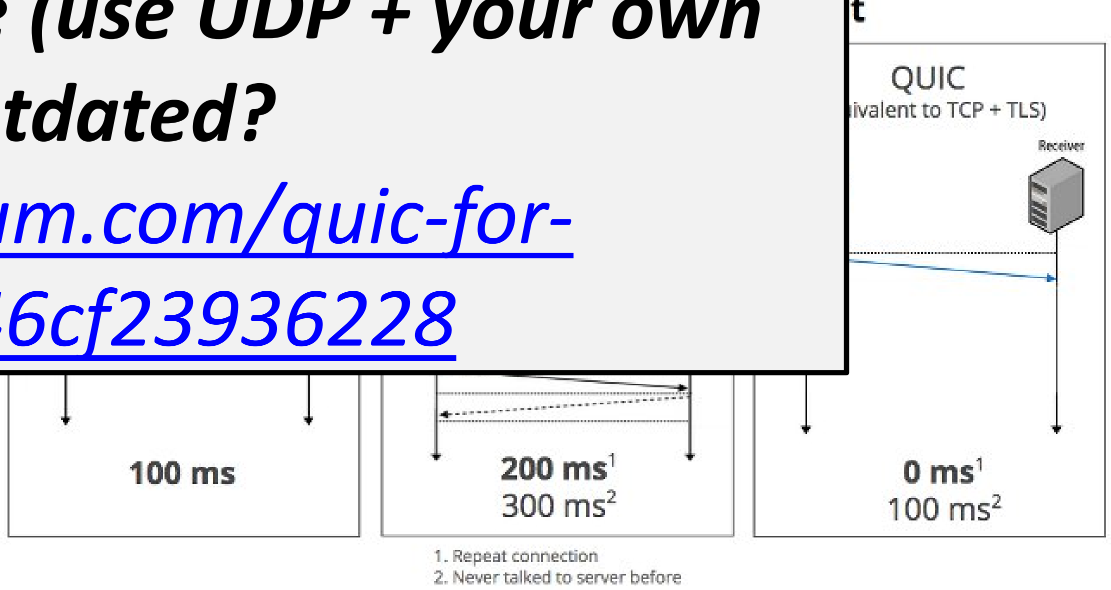


As we will see, QUIC might become a widespread standard, providing fast and configurable reliability. Hence, is it a good protocol for game networking? Is the content of this course (use UDP + your own reliability) outdated?

[*https://daposto.medium.com/quic-for-gamenetworking-46cf23936228*](https://daposto.medium.com/quic-for-gamenetworking-46cf23936228)



TLS and QUIC

It is important for on-line games because

- You may need to implement some secure connections (to log in, handle payments, etc.)
- Besides, gaming is progressively migrating to more secure paradigms thanks to the fast and configurable modes of DTLS and QUIC, not only encrypting payments, but also other messages (commands in peer-to-peer games)

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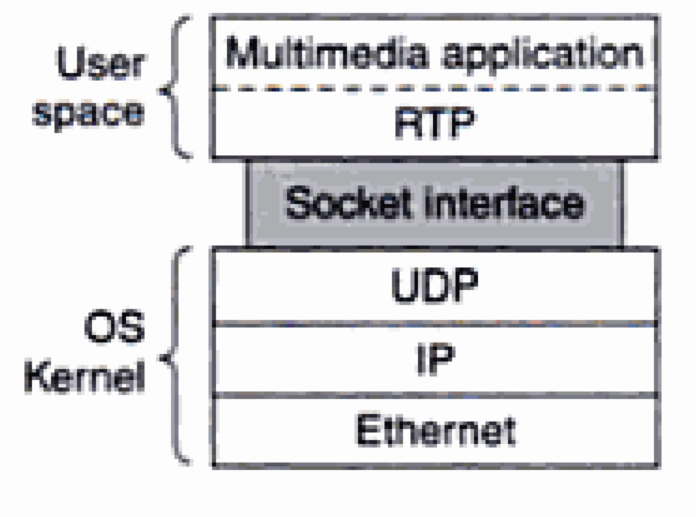
Protocols for streaming: RTP and WebRTC

RTP: Real-Time Transport Protocol



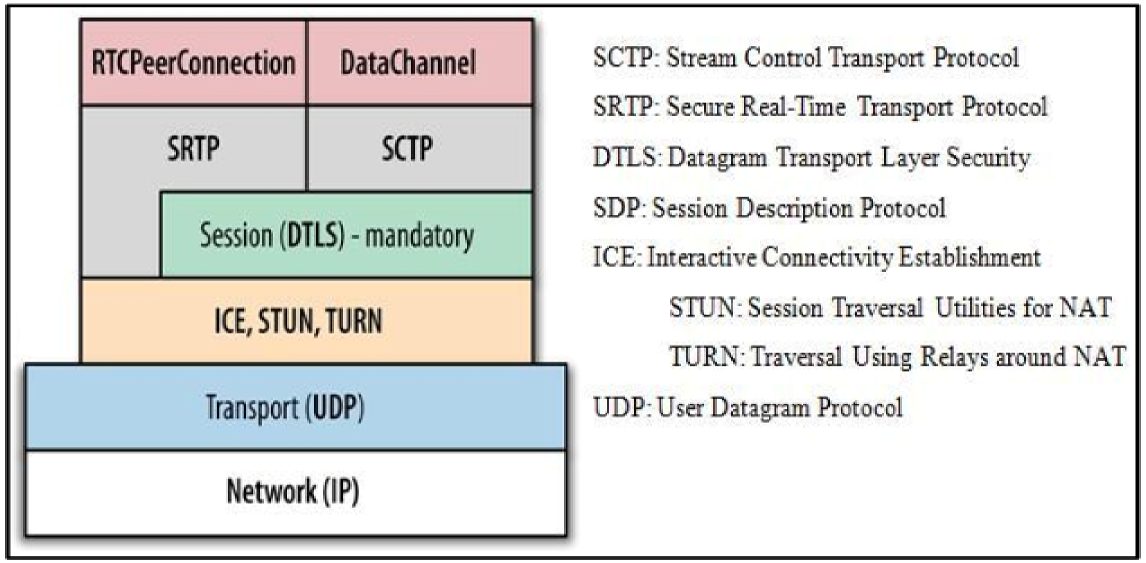
Layer: Application

- Application for delivery of audio and video over IP networks in real time.
- Typically used over UDP.
- As such, RTP implements mechanisms to react to losses and out-of-order.
- It has some complementary protocols:
  - **RTP Control Protocol (RTCP)** which helps with the network monitoring and synchronization of streams
  - **Secure RTP (SRTP)** which provides security for the RTP streams (encryption, authentication, integrity, etc.)

WebRTC: Web Real-Time Communication

WebRTC: Application/Session

- WebRTC is not really a protocol, but rather an API, to implement streaming between devices and web browsers effortlessly
- Uses a set of different protocols
  - UDP for quick response
  - ICE for establishing connections without problems
  - DTLS for quick security when establishing a connection
  - SRTP for secure streaming
- Open-source by Google, has the support from Apple, Microsoft, Mozilla and Opera among many others



RTP and WebRTC

It is important for on-line games because

- Cloud gaming is a possibility that is there and gaining traction lately, and uses these protocols

<table>
<tr><th></th><th colspan="2">Stadia</th><th>GeForce Now</th><th>PS Now</th></tr>
<tr><td>Streaming</td><td colspan="2">RTP (and RTCP)</td><td>RTP</td><td>Custom (UDP)</td></tr>
<tr><td>Player's input</td><td rowspan="4">WebRTC</td><td>DTLS</td><td>Custom (UDP)</td><td>Custom (UDP)</td></tr>
<tr><td>Session setup</td><td>DTLS, STUN</td><td>TLS</td><td>Custom (UDP)</td></tr>
<tr><td>Network Testing</td><td>RTP</td><td>Iperf-like</td><td>Custom (UDP)</td></tr>
<tr><td>Video Codec</td><td>H.264, VP9</td><td>H.264</td><td>-</td></tr>
</table>

Andrea Di Domenico, Gianluca Perna, Martino Trevisan, Luca Vassio and Danilo Giordano. A Network Analysis on Cloud Gaming: Stadia, GeForce Now and PSNow. MDPI Network (2021).

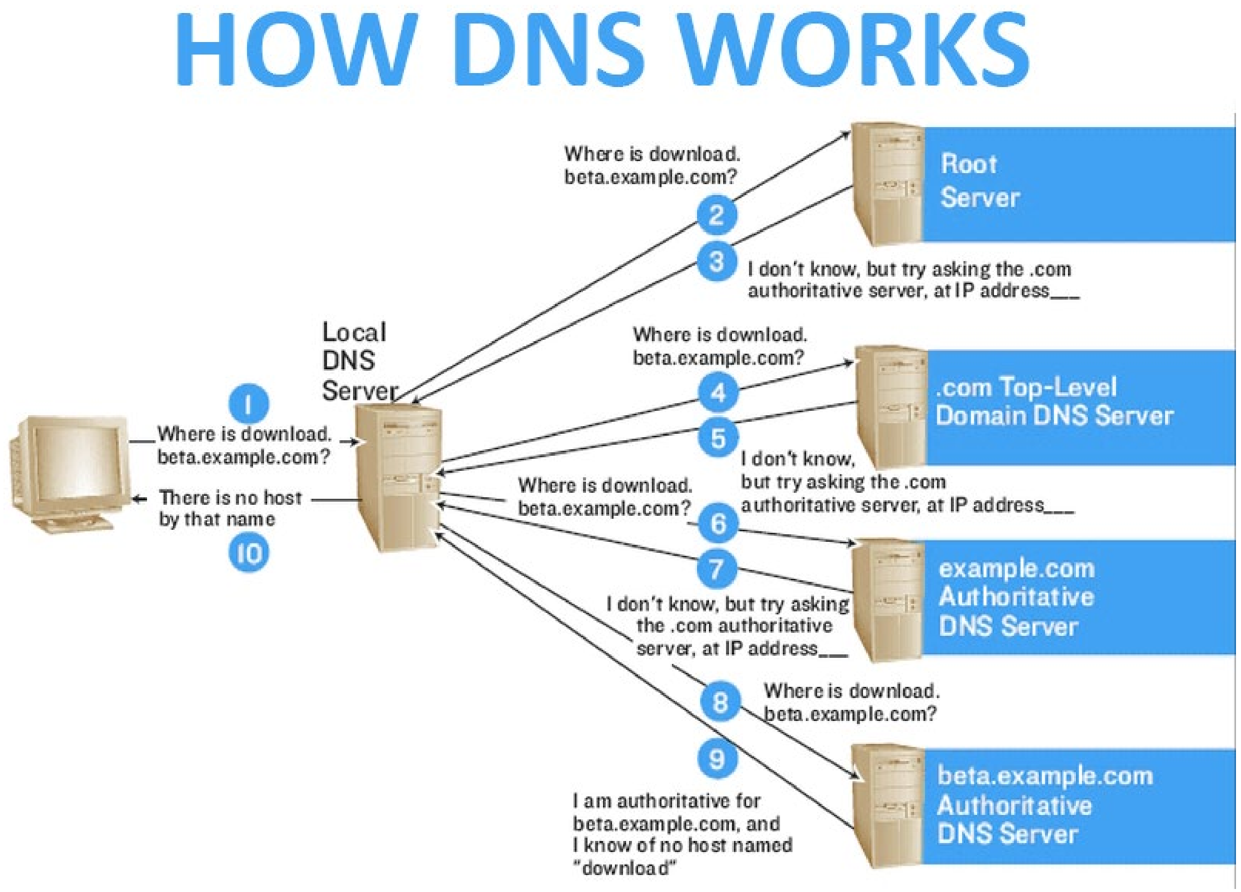
- Multiplayer AR/VR will require a streaming approach to work, and people are talking about WebRTC: https://www.liveswitch.io/blog/webrtc-multiparty-vr-applications

For other fundamental stuff: DNS and HTTP

DNS: Domain Name System

Layer: Application

- Allows to obtain the IP address associated to a domain name.
- From root (domain) to children (subdomains)
- Client-server architecture; that's why it is an application layer protocol



DNS: Domain Name System

Layer: Application

- On top of which protocols does DNS work?
- This has changed over time, to provide more reliability and security
  - UDP vs TCP
  - Adding TLS
  - Using QUIC

DNS over UDP/53 (Do53)

UDP reserves port number 53 for servers listening to queries.[5] Such queries consist of a clear-text request sent in a single UDP packet from the client, responded to with a clear-text reply sent in a single UDP packet from the server. When the length of the answer exceeds 512 bytes and both client and server support Extension Mechanisms for DNS (EDNS), larger UDP packets may be used.[42] Use of DNS over UDP is limited by, among other things, its lack of transport-layer encryption, authentication, reliable delivery, and message length.

DNS over TCP/53 (Do53/TCP)

In 1989, RFC 1123 specified optional Transmission Control Protocol (TCP) transport for DNS queries, replies and, particularly, zone transfers. Via fragmentation of long replies, TCP allows longer responses, reliable delivery, and re-use of long-lived connections between clients and servers.

DNS over TLS (DoT)

DNS over TLS emerged as an IETF standard for encrypted DNS in 2016, utilizing Transport Layer Security (TLS) to protect the entire connection, rather than just the DNS payload. DoT servers listen on TCP port 853. RFC 7858 specifies that opportunistic encryption and authenticated encryption may be supported, but did not make either server or client authentication mandatory.

DNS over HTTPS (DoH)

DNS over HTTPS was developed as a competing standard for DNS query transport in 2018, tunneling DNS query data over HTTPS, which transports HTTP over TLS. DoH was promoted as a more web-friendly alternative to DNS since, like DNSCrypt, it uses TCP port 443, and thus looks similar to web traffic, though they are easily differentiable in practice.[43]

Oblivious DNS (ODNS) and Oblivious DoH (ODoH)

Oblivious DNS (ODNS) was invented and implemented by researchers at Princeton University and the University of Chicago as an extension to unencrypted DNS,[44] before DoH was standardized and widely deployed. Apple and Cloudflare subsequently deployed the technology in the context of DoH, as Oblivious DoH (ODoH).[45] ODoH combines ingress/egress separation (invented in ODNS) with DoH's HTTPS tunneling and TLS transport-layer encryption in a single protocol.[46]

security

DNS over TOR

DNS may be run over virtual private networks (VPNs) and tunneling protocols. A use which has become common since 2019 to warrant its own frequently used acronym is DNS over Tor. The privacy gains of Oblivious DNS can be garnered through the use of the preexisting Tor network of ingress and egress nodes, paired with the transport-layer encryption provided by TLS.[47]

DNS over QUIC (DoQ)

RFC 9250 by the Internet Engineering Task Force describes DNS over QUIC.

DNSCrypt

DNS: Domain Name System

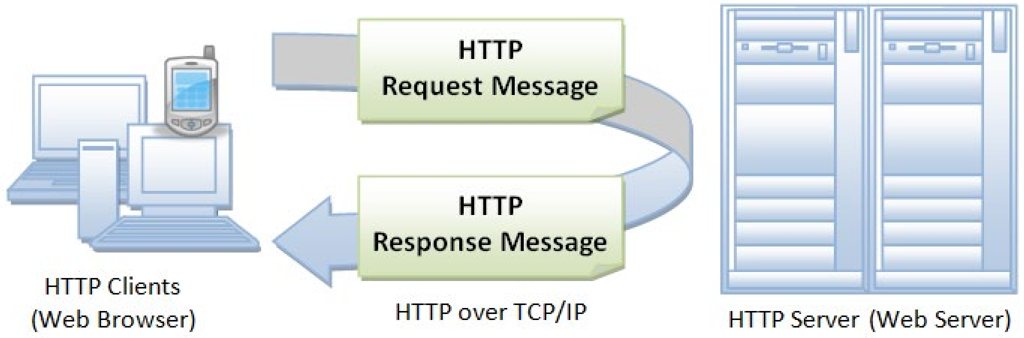
It is important for on-line games because

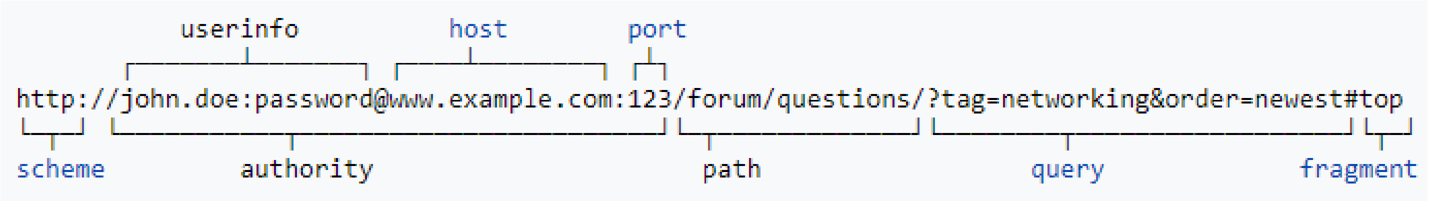
- You do not want your server to be identifiable by a hardcoded IP, which will possibly change over time if you want to migrate servers or your Internet Service Provider (ISP) causes you problems
- Sockets do not admit domain names. Therefore, clients may want to first resolve the IP address that they need to connect to the server through the socket

HTTP: HyperText Transfer Protocol

Layer: Application

- Implements distributed, collaborative, hypermedia information systems
- Client-server architecture, that's why it's an application layer
- Defines hypertext, hyperlinks as resources with a given format (Uniform Resource Locator, URL)

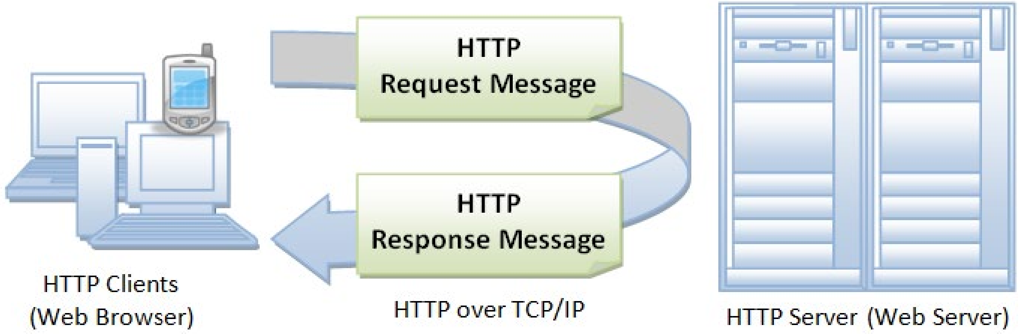




HTTP: HyperText Transfer Protocol

Layer: Application

- It has a specific structure of client requests with different methods
  - GET
  - HEAD
  - POST
  - ...

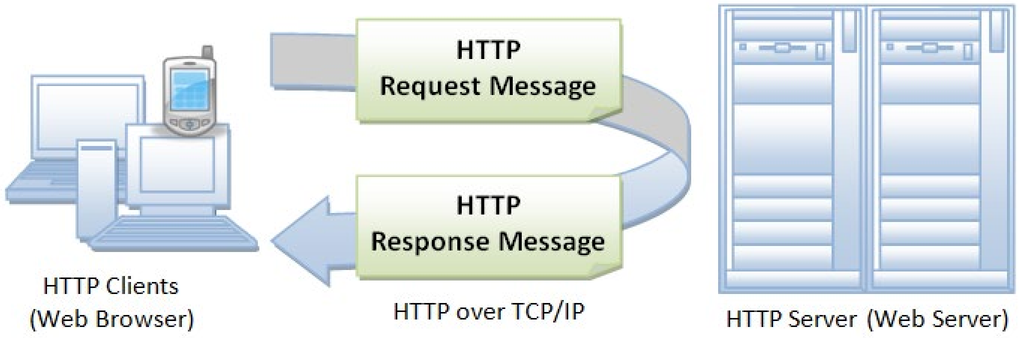



```
GET / HTTP/1.1
Host: www.example.com
```

HTTP: HyperText Transfer Protocol

Layer: Application

- It has a specific structure of server replies with different codes
  - 200 (OK)
  - 401 (not authorized)
  - **404 (not found)**
  - 500 (internal server error)
  - ...



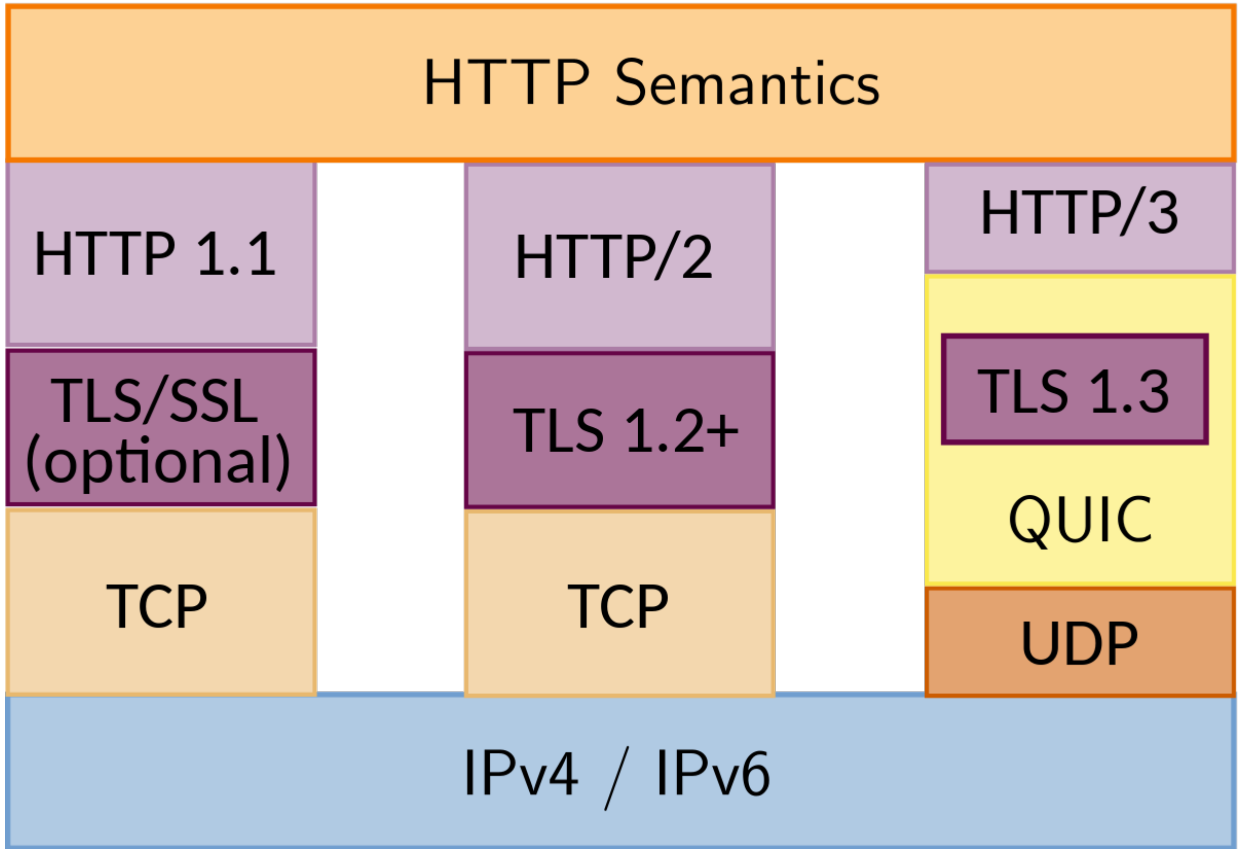

```
HTTP/1.1 200 OK
Date: Mon, 23 May 2005 22:38:34 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 138
Last-Modified: Wed, 08 Jan 2003 23:11:55 GMT
Server: Apache/1.3.3.7 (Unix) (Red-Hat/Linux)
ETag: "3f80f-1b6-3e1cb03b"
Accept-Ranges: bytes
Connection: close

<html>
  <head>
    <title>An Example Page</title>
  </head>
  <body>
    <p>Hello World, this is a very simple HTML document.</p>
  </body>
</html>
```

HTTP: HyperText Transfer Protocol

Layer: Application

- HTTP has been evolving to improve performance, reliability, and security
  - Pipelining requests and responses
  - Adding server-side push
  - Adopting QUIC as a faster and configurable alternative to TCP+TLS.



HTTP: HyperText Transfer Protocol

It is important for on-line games because

- It gives you yet another example of client-server communication protocol and how it has evolved over time, adopting new techniques to increase performance, reliability, and security.
- HTTP is at the foundation of the Internet and many things, from connections to websites to things like the REST API are based on HTTP.

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The Protocol Stack

2.3 Other relevant protocols

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Xarxes i Jocs On-line – 2025-2026
Grau en Disseny i Desenvolupament de Videojocs

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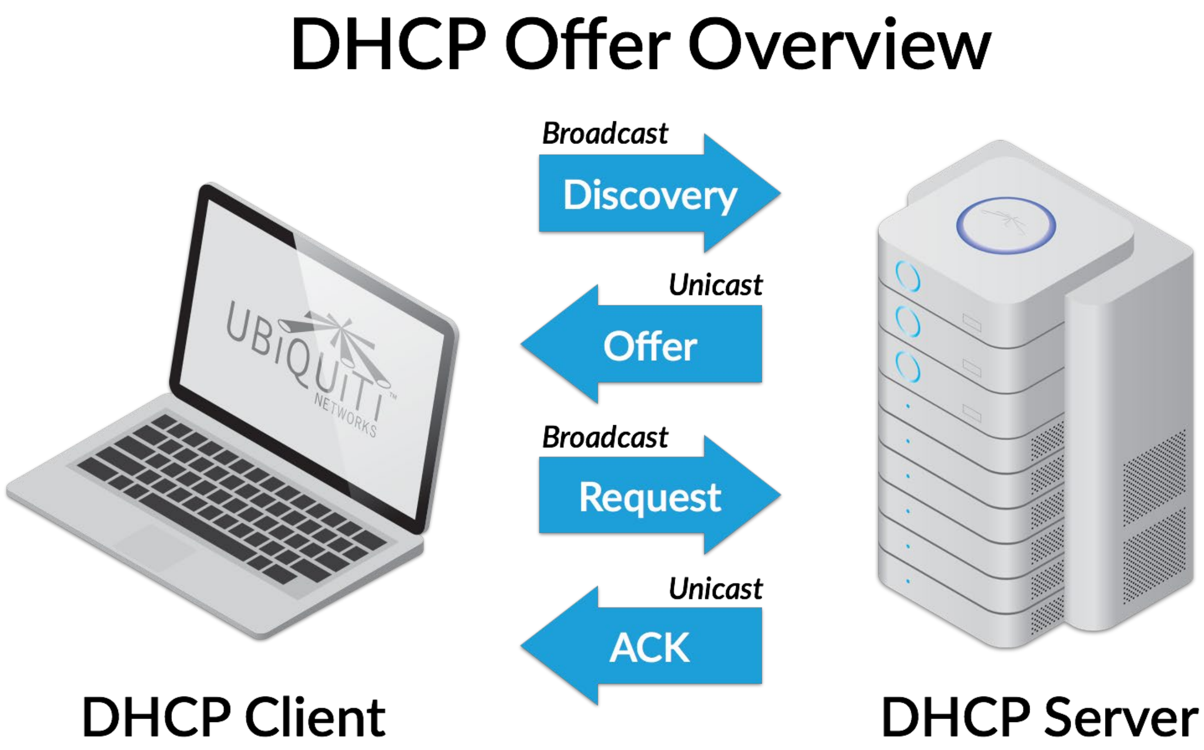
Extra material: Dynamic Host Configuration Protocol (DHCP)

DHCP: Dynamic Host Configuration Protocol

Layer: Application

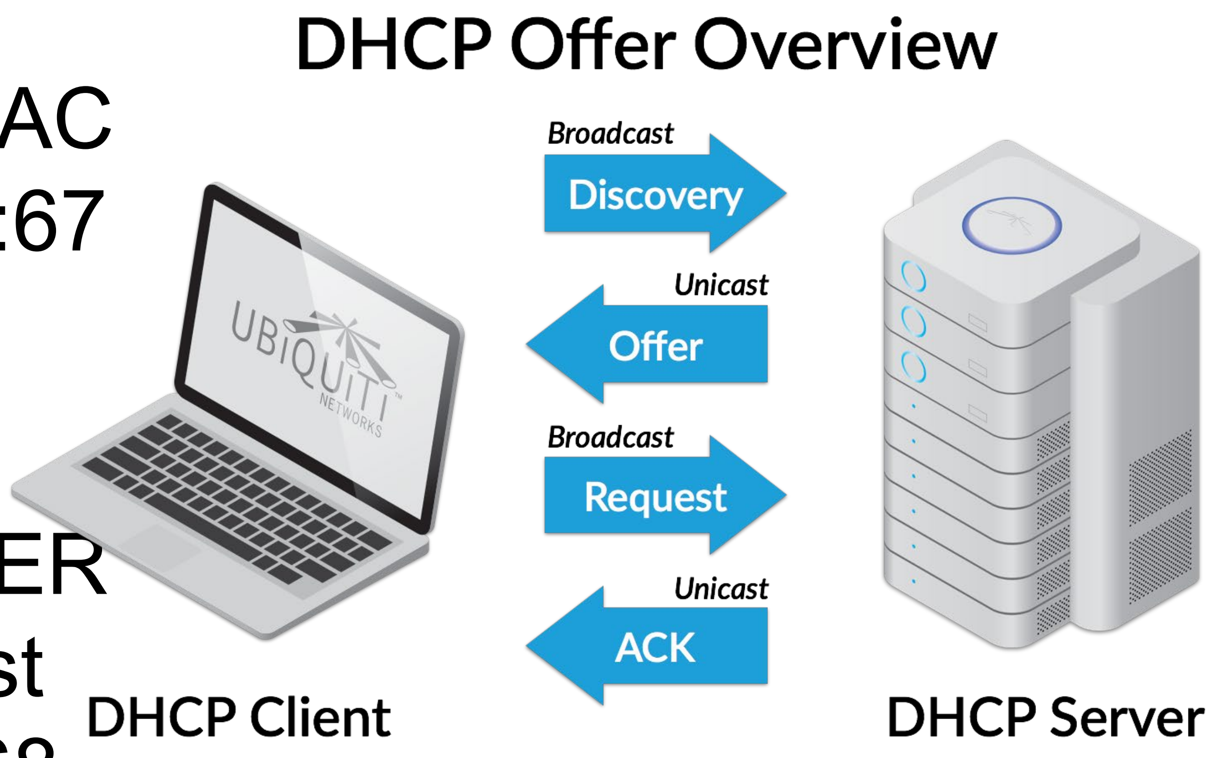
- You have a limited range of IP addresses, for a number of devices that is larger.
- DHCP allows to obtain a valid IP address in a non-static way (with exceptions) from a pool.
- Client-server architecture.



DHCP: Dynamic Host Configuration Protocol

Process (I):

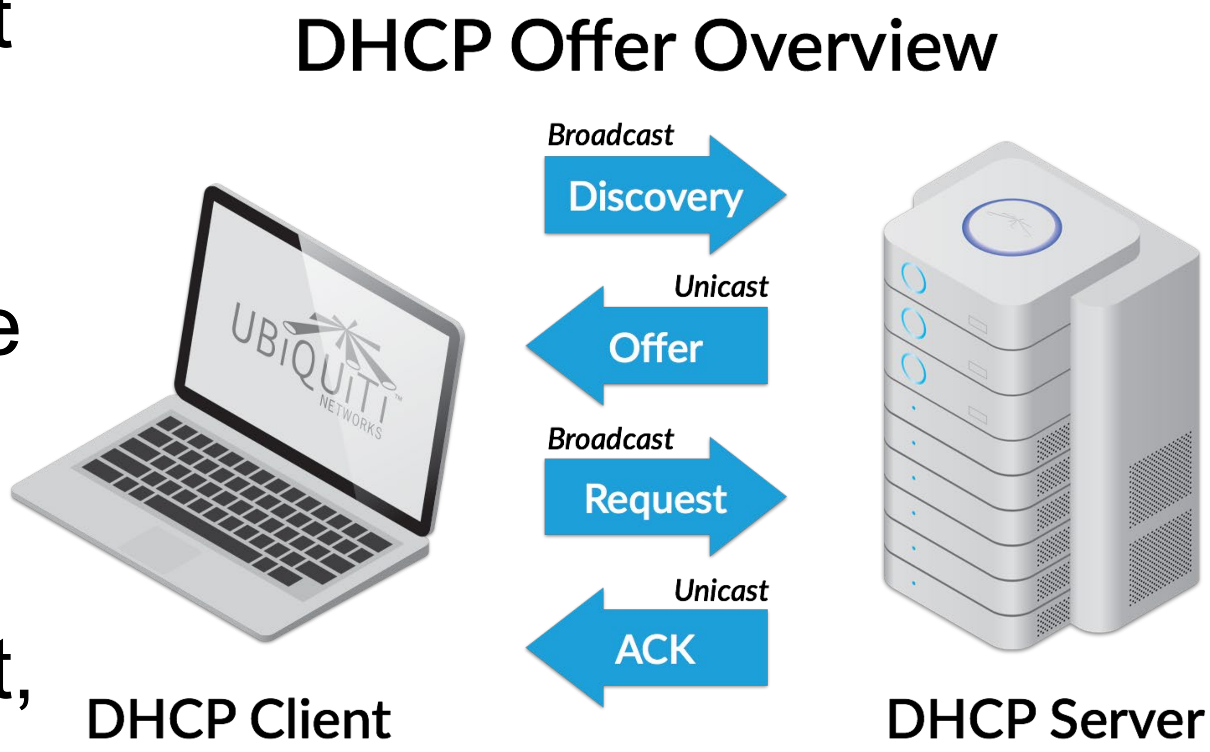
- **Discovery:** client sends DHCPDISCOVER with its MAC address to 255.255.255.255:67 (broadcast)
- **Offer:** server offers an IP address sending DHCPOFFER (+MAC address) to broadcast address of the subnet, port 68. All nodes check if it's for them



DHCP: Dynamic Host Configuration Protocol

Process (II):

- **Request:** node with the right MAC accepts the offer, and sends DHCPREQUEST in broadcast to all, reaching the server
- **ACK:** server acknowledges the request and, at that point, the client has its new IP assigned.

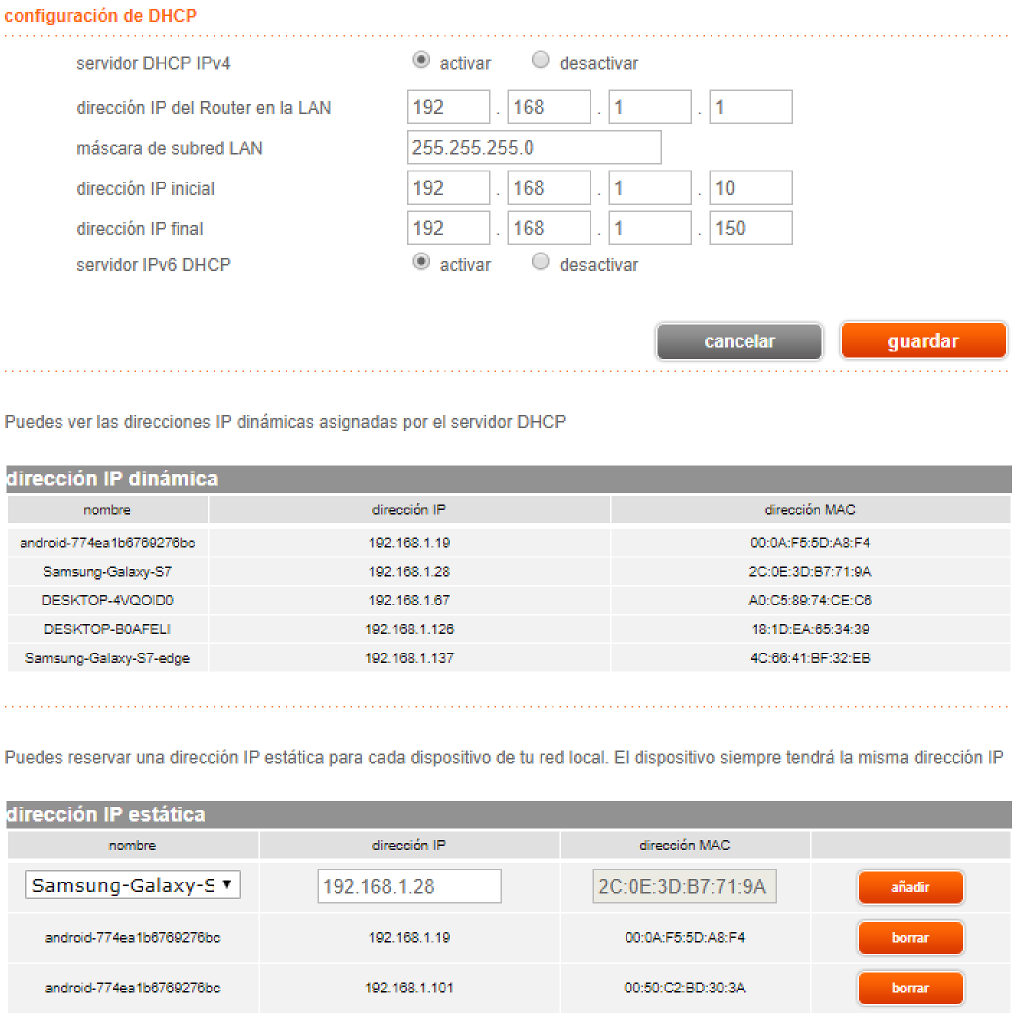


DHCP: Dynamic Host Configuration Protocol

CITM

Layer: Application

- Example: private in your LAN (server is your router)
- Example: public in your WAN (server given by the operator)


configuración de DHCP

servidor DHCP IPv4 — activar / desactivar

dirección IP del Router en la LAN — 192 . 168 . 1 . 1

máscara de subred LAN — 255.255.255.0

dirección IP inicial — 192 . 168 . 1 . 10

dirección IP final — 192 . 168 . 1 . 150

servidor IPv6 DHCP — activar / desactivar

cancelar | guardar

Puedes ver las direcciones IP dinámicas asignadas por el servidor DHCP

dirección IP dinámica

nombre	dirección IP	dirección MAC
android-774ea1b6769276bc	192.168.1.19	00:0A:F5:5D:A8:F4
Samsung-Galaxy-S7	192.168.1.28	2C:0E:3D:B7:71:9A
DESKTOP-4VQOID0	192.168.1.67	A0:C5:89:74:CE:C6
DESKTOP-B0AFELI	192.168.1.126	18:1D:EA:65:34:39
Samsung-Galaxy-S7-edge	192.168.1.137	4C:66:41:BF:32:EB

Puedes reservar una dirección IP estática para cada dispositivo de tu red local. El dispositivo siempre tendrá la misma dirección IP

dirección IP estática

nombre	dirección IP	dirección MAC	
Samsung-Galaxy-S	192.168.1.28	2C:0E:3D:B7:71:9A	añadir
android-774ea1b6769276bc	192.168.1.19	00:0A:F5:5D:A8:F4	borrar
android-774ea1b6769276bc	192.168.1.101	00:50:C2:BD:30:3A	borrar